

Global Aviation Disruption Assessment 2026

Technical Documentation

Implementation reference for data preparation, PostgreSQL analysis, the completed Power BI model, and local reproduction.

Scenario note: This is a simulated ICAO client assignment, not an official ICAO commission or endorsement.

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1 System overview

The implemented system is a static batch workflow: seven raw CSV files are validated and cleaned by Python, seven UTF-8 outputs are loaded into PostgreSQL, SQL scripts and views provide analytical interfaces, and a Power BI Import model presents four report pages. Source populations remain separate throughout the pipeline.

The database is aviation_bi and the schema is aviation. PostgreSQL contains seven source-aligned analytical tables and nine views. No primary or foreign keys are declared. Power BI relationships are semantic-model relationships and do not imply database foreign keys.

2 Repository architecture and execution order

Layer	Location and responsibility
Raw and clean data	data/raw holds the original snapshot; data/clean holds the seven reproducible UTF-8 outputs.
Python	python/etl contains seven numbered scripts; python/utils/helpers.py contains shared validation and file handling.
Database setup	sql/01_database creates the database, schema, and seven analytical tables.
Loading and checks	sql/02_loading contains the psql loader; sql/03_validation checks loads and view reconciliation.
Analysis and views	sql/04_analysis contains six analytical scripts; sql/05_views contains nine view definitions.
Power BI	powerbi contains the PBIP entry point, report definition, and semantic-model definition.
Published evidence	documentation contains four reader-facing documents; visuals contains diagrams and final report images.

Run Python scripts in numeric order. For a new local database, run database creation, schema creation, table creation, loading, view creation, validation, and analysis in that order. View creation precedes validation because validation references reporting views. Refresh Power BI after the database is ready.

3 End to end lineage

airline_losses.csv → numbered Python ETL → airline_losses_clean.csv → aviation.airline_losses. Summary loss, cancellations, reroutes, revenue-loss percentage.

airline_losses_estimate.csv → numbered Python ETL → airline_losses_estimate_clean.csv → aviation.airline_losses_estimate. Daily loss and estimated passengers, fuel, cancellations, reroutes.

airport_disruptions.csv → numbered Python ETL → airport_disruptions_clean.csv → aviation.airport_disruptions. Airport flights affected, duration, severity, geography.

airspace_closures.csv → numbered Python ETL → airspace_closures_clean.csv → aviation.airspace_closures. Closure count, hours, and reported flight exposure.

conflict_events.csv → numbered Python ETL → conflict_events_clean.csv → aviation.conflict_events. Event timeline, military records, reported severity.

flight_cancellations.csv → numbered Python ETL → flight_cancellations_clean.csv → aviation.flight_cancellations. Cancellation records and affected passengers.

flight_reroutes.csv → numbered Python ETL → flight_reroutes_clean.csv → aviation.flight_reroutes. Reroute records, distance, fuel, and delay.

Financial tables feed vw_airline_loss_summary and vw_airline_loss_estimate separately. Airport and airspace tables feed their detail interfaces. The regional view combines regional aggregates as separate columns. The route view combines daily country-pair aggregates while keeping cancellation and reroute measures separate. The geopolitical view aligns daily records to a continuous calendar.

Project Architecture.png shows the workflow. ERD -1.png distinguishes the PostgreSQL structure from the Power BI semantic model. Source-to-view lineage is different from a semantic relationship.

4 Data source and population definitions

Dataset	Rows and grain	Classification and date scope
airline_losses	70; One row per airline	Undated modeled summary; Undated
airline_losses_estimate	35; One row per airline	Undated modeled estimate; Undated
airport_disruptions	227; One disruption record	Simulated airport detail; 2026-02-27 to 2026-06-07
airspace_closures	84; One closure record	Simulated closure detail; 2026-02-28 to 2026-06-07
conflict_events	82; One contextual event	Simulated event context; 2025-12-15 to 2026-06-08
flight_cancellations	2,200; One cancellation record	Simulated operational detail; 2026-02-27 to 2026-06-07
flight_reroutes	1,500; One reroute record	Simulated operational detail; 2026-02-28 to 2026-06-07

Source: Global Civil Aviation Disruption 2026 Iran-US War by zkskhurram, <https://www.kaggle.com/datasets/zkskhurram/global-civil-aviation-disruption2026-iranus-war>. All seven files were downloaded in the first week of August 2026. The raw and cleaned datasets are licensed under Creative Commons Attribution-ShareAlike 4.0 International (CC BY-SA 4.0), <https://creativecommons.org/licenses/by-sa/4.0/>. Raw files are unchanged. Preparation includes text trimming, date formatting, three airline-name aliases, closure timestamp formatting and duration validation and conflict-location parsing, as detailed in Section 5. Modified data remains

under CC BY-SA 4.0. Repository code is licensed under MIT; that code license does not cover the source or cleaned datasets. No endorsement by the dataset creator is implied.

4.1 Reconciliation rules

The summary has 6,030 cancellations and 4,539 reroutes; detailed records have 2,200 and 1,500. Summary and estimate populations contain 70 and 35 airlines, with 32 shared and 73 in the union. These observations do not establish a common scope. Financial and detailed totals are not reconciled by adding, deleting, or rewriting records.

A date-airline-flight number tuple is only a diagnostic candidate, not a declared unique key. Airport and airspace exposure cannot be deduplicated across sources. Country strings are source labels; no external country-standard mapping is applied.

5 Python implementation

5.1 Runtime and shared helpers

requirements.txt declares pandas==3.0.3. pathlib and sys are Python standard-library modules. Python 3.14.6 is the documented project environment. Each ETL script has a main entry point, resolves helpers relative to its file, validates expected column order, and writes only its corresponding clean CSV.

Helper	Implemented responsibility
load_raw_csv	Load one UTF-8 source file and reject unexpected schema drift.
trim_text	Trim leading and trailing whitespace without changing null values.
standardize_airline_names	Apply only the three documented aliases used across financial tables.
parse_dates	Parse required date/timestamp fields strictly.
require_values	Reject nulls in fields that define the dataset's documented grain.
require_nonnegative	Reject negative values in count, duration, distance and cost fields.
require_range	Reject values outside an inclusive documented range.
require_categories	Reject undocumented categorical values while allowing schema drift to surface.
duplicate_count	Return duplicate-row or repeated-candidate-key count for reporting.
assert_row_preservation	Ensure cleaning has neither removed nor added source records.
save_clean_csv	Write a deterministic UTF-8 clean file with ISO dates and timestamps.
print_validation_summary	Print a concise, comparable result for each standalone ETL run.

The repository root is resolved from helpers.py. Raw input is read as UTF-8. Text trimming preserves null values. Required fields, allowed categories, numeric ranges, nonnegative values, strict dates, and row preservation are checked where each ETL script calls the helper. Output dates use YYYY-MM-DD and timestamps use YYYY-MM-DD HH:MM:SS.

5.2 Dataset transformations

5.2.1 airline losses

Trim text; require every field; validate nonnegative loss and disruption counts, revenue percentage from 0 to 100, and four airline types. Preserve financial values.

Input: data/raw/airline_losses.csv. Output: data/clean/airline_losses_clean.csv. Raw and clean rows: 70. Input columns: 8; output columns: 8.

5.2.2 airline losses estimate

Trim text and apply PIA → Pakistan International Airlines, Saudia → Saudi Arabian Airlines, Swiss International → Swiss International Air Lines. Require all fields and nonnegative numeric values.

Input: data/raw/airline_losses_estimate.csv. Output: data/clean/airline_losses_estimate_clean.csv. Raw and clean rows: 35. Input columns: 7; output columns: 7.

5.2.3 airport disruptions

Trim text, parse date, require fields, validate nonnegative counts and hours, and reject IATA codes mapping to multiple airport names or countries.

Input: data/raw/airport_disruptions.csv. Output: data/clean/airport_disruptions_clean.csv. Raw and clean rows: 227. Input columns: 9; output columns: 9.

5.2.4 airspace closures

Trim text and parse both timestamps. Require fields, reject negative measures and end-before-start intervals, and reject duration differences above 0.011 hours. Duration is validated, not overwritten.

Input: data/raw/airspace_closures.csv. Output: data/clean/airspace_closures_clean.csv. Raw and clean rows: 84. Input columns: 8; output columns: 8.

5.2.5 conflict events

Trim text and parse date. Copy location to source_location, split on the first comma to obtain location_name and country, trim both components, and preserve null country. Require exactly 42 unresolved countries for this fixed snapshot.

Input: data/raw/conflict_events.csv. Output: data/clean/conflict_events_clean.csv. Raw and clean rows: 82. Input columns: 6; output columns: 8.

5.2.6 flight cancellations

Trim text, parse date, require fields, validate nonnegative passengers, and report repeated flight-number rows without deleting them.

Input: data/raw/flight_cancellations.csv. Output: data/clean/flight_cancellations_clean.csv. Raw and clean rows: 2,200. Input columns: 12; output columns: 12.

5.2.7 flight reroutes

Trim text, parse date, require fields, validate nonnegative numeric values, and reject new distance minus original distance minus additional distance differences above 0.001 km. Report repeated flight numbers without deleting them.

Input: data/raw/flight_reroutes.csv. Output: data/clean/flight_reroutes_clean.csv. Raw and clean rows: 1,500. Input columns: 16; output columns: 16.

Exact duplicate checks are reported; the helper does not automatically remove duplicate rows. Schema errors, invalid dates, required nulls, range violations, and formula mismatches raise exceptions before the clean output is saved. Each script is standalone; there is no all-seven-file transaction.

6 PostgreSQL implementation

6.1 Schema and constraints

The database uses UTF-8. The source-aligned tables preserve each dataset grain. Required fields are NOT NULL except conflict_events.country. Numeric checks enforce nonnegative values. Revenue-loss percentage is bounded from 0 to 100, IATA code length is three, closure end must not precede start, and reroute distance arithmetic is checked.

Airport and conflict severity vocabularies are different and have separate checks. Airline type uses Cargo, Flag Carrier, Low Cost, and Private. No invented cross-table identifiers or relational keys are added.

6.2 Loading and failure behavior

01_load_data.psql is a psql client script, not ordinary Query Tool SQL. It sets ON_ERROR_STOP and UTF8 client encoding, begins a transaction, truncates the seven project tables, and uses client-side \copy with explicit columns and repository-relative clean-file paths. The loader checks all seven row counts before COMMIT. A failing command or count assertion prevents a committed partial load; close or roll back the failed session before retrying.

03_create_tables.sql is a destructive local rebuild script: DROP TABLE CASCADE removes project tables and dependent views before recreating them. Run it only for deliberate setup or rebuilding. CREATE DATABASE must run outside a transaction against the postgres maintenance database.

7 Validation responsibilities

Layer	Checks owned by the layer
Python	Raw schema, required fields, date parsing, numeric and category rules, formula checks, repeated-record diagnostics, and raw-to-clean row preservation.
SQL	Loaded row counts, material totals, preserved source locations, financial coverage, view row/aggregate preservation, and daily calendar coverage.
Power BI	Population-specific measures, relationship and filter rules, report labels, source-aligned totals, and appropriate date behavior.

7.1 File control totals

Control	Value	Source population
Detailed cancellation records	2200	flight_cancellations
Cancellation passengers	575710	flight_cancellations
Detailed reroute records	1500	flight_reroutes
Additional distance km	2201390	flight_reroutes
Extra fuel cost USD	11,658,572.59	flight_reroutes
Reroute delay hours	4,706.30	flight_reroutes
Modeled summary loss USD	21,414,330,000	airline_losses
Modeled summary cancellations	6030	airline_losses
Modeled summary reroutes	4539	airline_losses
Estimated daily loss USD	48,840,000	airline_losses_estimate
Estimated fuel cost USD	27,606,000	airline_losses_estimate
Estimated passengers	119570	airline_losses_estimate
Estimated reroutes	754	airline_losses_estimate
Estimated cancellations	195	airline_losses_estimate
Airport flights affected	35844	airport_disruptions
Airport disruption hours	8,218.80	airport_disruptions
Airspace flights affected	35502	airspace_closures
Airspace closure hours	7014	airspace_closures
Conflict records	82	conflict_events
Military records	18	conflict_events

These controls can be independently reproduced from the clean files and compared with SQL output and unfiltered report values. File controls are not a substitute for checking a new local database load. The current dashboard export agrees with its displayed rounded totals.

8 SQL analysis catalog

Analysis file	Implemented output	Questions
01_airport_disruption_analysis.sql	Region, country, airport totals, longest records, and daily regional airport exposure.	Q12 to Q16
02_airline_financial_scenario_analysis.sql	Airline, region, and airline-type modeled summary rankings.	Q19 to Q21
03_airline_estimate_analysis.sql	Daily estimates, within-estimate ratios, totals, and unmatched airline coverage.	Q22 to Q23
04_geopolitical_temporal_analysis.sql	Military-event versus other-date means in shared coverage and wider contextual timeline.	Q24
05_regional_route_analysis.sql	Separate regional exposures and daily country-pair route aggregates.	Q09 to Q11; Q18
06_executive_reconciliation.sql	Population-specific headline counts and material totals.	Q01 Q02 Q05 to Q08

Q03 and Q04 are answered by the report daily trend and cancellation-reason visual using detailed cancellation facts. Q17 is answered by the airspace-country report ranking. Additional grouping used in the case study is a descriptive aggregation of the same facts, not a new repository query or a new business population.

Analysis 04 uses 28 February to 7 June 2026, comprising 12 military-event dates and 88 other dates. The wider daily view spans 15 December 2025 to 17 June 2026. The view fills absent daily counts with zero; analysis 04 masks out-of-coverage operational values as NULL in its contextual output. Source date bounds do not establish complete real-world observation.

9 SQL view catalog

View	Grain	Role and use
vw_airline_loss_summary	One summary airline	Authoritative summary; imported
vw_airline_loss_estimate	One estimate airline	Authoritative estimate; imported
vw_airport_disruption	One airport disruption record	Supporting detail; imported
vw_airspace_closure	One closure record	Supporting detail; imported
vw_geopolitical_daily_impact	One calendar date	Supporting contextual baseline; imported
vw_regional_vulnerability	One region	Separate source aggregates; imported

View	Grain	Role and use
vw_route_passenger_impact	Date plus origin/destination region and country	Supporting route aggregates; imported
vw_airline_operational_financial	One union airline	Deprecated compatibility; not imported; used by SQL validation
vw_executive_kpi	One summary row	Deprecated compatibility; not imported

The deprecated financial view uses a full outer join and presence flags to preserve both financial populations. Its ratios use estimate fields only. Neither deprecated view feeds the final Power BI report. Their legacy SQL comments refer to earlier phases; these comments do not describe current model migration status.

The geopolitical view counts closures across inclusive start-to-end calendar dates, not prorated hours. The regional view keeps airport, closure, and modeled-loss columns distinct. The route view full-joins pre-aggregated facts at the shared date and geography grain, preventing raw fact-to-fact multiplication.

10 Final Power BI implementation

10.1 Project and report pages

The PBIP project uses Import mode and PostgreSQL.Database("localhost", "aviation_bi"). The semantic model contains 22 tables, 33 explicit measures, and 35 relationships. Four report pages contain 76 visual definition files, including decorative and text elements.

Page	Purpose
Executive Overview	Detailed operations, financial overview, airport concentration, and event context.
Modeled Financial Scenarios	Separate summary and estimate measures, airline rankings, and within-estimate ratios.
Geopolitical & Operational Disruption	Date trends, cancellation reasons, airports, airspace hours, and military records.
Regional Risk & Corridor Vulnerability	Directional corridor activity, distance burden, and separate regional exposures.

10.2 Imported sources and Power Query

Model table	PostgreSQL source
aviation conflict_events	aviation.conflict_events
aviation flight_cancellations	aviation.flight_cancellations

Model table	PostgreSQL source
aviation flight_reroutes	aviation.flight_reroutes
aviation vw_airline_loss_estimate	aviation.vw_airline_loss_estimate
aviation vw_airline_loss_summary	aviation.vw_airline_loss_summary
aviation vw_airport_disruption	aviation.vw_airport_disruption
aviation vw_airspace_closure	aviation.vw_airspace_closure
aviation vw_geopolitical_daily_impact	aviation.vw_geopolitical_daily_impact
aviation vw_regional_vulnerability	aviation.vw_regional_vulnerability
aviation vw_route_passenger_impact	aviation.vw_route_passenger_impact

Power Query selects each object from schema aviation and imports its rows. The closure table also provides date-only start and end columns for calendar relationships. Route columns support a directional corridor label. Appendix C records the exact saved Power Query definitions for all imported sources.

10.3 Dimensions and relationships

Ten calculated dimensions cover date, airline, entity country, entity region, origin country, destination country, origin region, destination region, airport severity, and conflict severity. Report Measures is a disconnected measure container; Operational Factor is a disconnected selector used to display airport and airspace exposure separately.

All 35 relationships use single-direction filtering from the one side to the many side. Thirty-four are active. The closure end-date relationship is inactive; closure start date is the active date role. Appendix B lists every relationship and state. There are no direct fact-to-fact relationships.

DimAirline filters summary, estimate, cancellation, and reroute facts. Entity country filters financial, airport, closure, and assigned-country conflict facts. Entity region filters summary, airport, closure, and regional aggregate facts. Origin and destination dimensions filter the operational and route facts separately. Entity geography is not route geography.

The estimate source has no region or airline type. Entity Region and Airline Type therefore do not filter it through a direct relationship. The financial page configures visual interactions to avoid implying that unsupported source attributes apply to the estimate visuals. DimCountry does not supply an invented country to unresolved conflict records.

10.4 Measures and financial calculation rules

All 33 explicit measures are stored in Report Measures and grouped by operational detail, modeled summary, modeled estimate, geographic exposure, and route analysis. Appendix A lists each saved DAX expression. Counts use COUNTROWS; additive fields use SUM. Estimated ratios use DIVIDE on estimate measures and return blank for a zero denominator.

Modeled Average Revenue Loss Pct divides the source percentage average by 100 and formats it as a percentage. It is an unweighted airline mean. Estimated passengers per estimated reroute describes

exposure in the estimate population, not occupancy. Estimated daily loss is not multiplied by the operational day count or added to summary loss.

10.5 Date coverage and report interpretation

DimDate is a continuous calendar calculated from the minimum and maximum dates across dated facts and closure start/end dates. It spans 15 December 2025 to 17 June 2026. It has no relationship to either undated financial source or the undated regional aggregate.

Cancellation and airport records cover 27 February to 7 June; reroute records cover 28 February to 7 June. Earlier calendar selections can return blank operational measures because those sources have no records there. The Executive Overview contains an operational-coverage note; a separate dynamic availability measure is not present in the saved model. Financial figures remain undated when a date is selected.

Airspace date filtering selects closure start dates and sums complete record durations. It does not prorate hours into the selected date interval. The inactive closure-end relationship is available for a future explicit end-date calculation but is not the default report date role.

The final dashboard PDF contains four unfiltered pages. Its headline counts, financial amounts, airport rankings, closure-country hours, and directional corridor counts agree with the source calculations after displayed rounding. A static export does not demonstrate interactive behavior; relationships, measures, and saved visual interactions define that behavior.

11 Local reproduction

11.1 Prerequisites

Use Python 3.14.6 with pandas 3.0.3, PostgreSQL with the psql client, and Power BI Desktop supporting PBIP and TMDL. Clone the full repository; keep the PBIP entry point beside its .Report and .SemanticModel folders. Use a dedicated local project database, not an unrelated database.

11.2 Python procedure

1. Open a terminal at the repository root.
2. Create an environment with `python -m venv .venv`.
3. Activate it with `.\venv\Scripts\Activate.ps1`.
4. Install dependencies with `python -m pip install -r requirements.txt`.
5. Run each numbered file under `python/etl` with `python` followed by its path, from 01 through 07.
6. Confirm the seven validation summaries and the clean outputs.

11.3 PostgreSQL procedure

The following commands use `postgres` as an example local administrator username. Substitute your own authorized username. The `-W` option prompts for a password; do not put it in a command, source file, or public document.

1. `psql -h localhost -U postgres -d postgres -W -v ON_ERROR_STOP=1 -f sql/01_database/01_create_database.sql`
2. `psql -h localhost -U postgres -d aviation_bi -W -v ON_ERROR_STOP=1 -f sql/01_database/02_create_schema.sql`

3. `psql -h localhost -U postgres -d aviation_bi -W -v ON_ERROR_STOP=1 -f sql/01_database/03_create_tables.sql`
4. `psql -h localhost -U postgres -d aviation_bi -W -f sql/02_loading/01_load_data.psql`
5. `psql -h localhost -U postgres -d aviation_bi -W -v ON_ERROR_STOP=1 -f sql/05_views/01_create_views.sql`
6. `psql -h localhost -U postgres -d aviation_bi -W -v ON_ERROR_STOP=1 -f sql/03_validation/01_validation.sql`

Run each of the six sql/04_analysis files with the same database connection and ON_ERROR_STOP setting. If aviation_bi already exists, skip database creation. Do not rerun table rebuild or truncate/load merely to view query results.

11.4 Power BI procedure

1. Open Global Aviation Disruption Assessment 2026 DASHBOARD.pbip from powerbi.
2. Confirm PostgreSQL is running and aviation_bi contains the expected tables and views.
3. In data source settings, use localhost and aviation_bi. Enter your local database username and password in the credential dialog.
4. Refresh the imported model. Review the headline totals against section 7 and test applicable date, entity, origin, destination, and severity filters.
5. Save the project. Keep credentials and .pbi local state out of source control.

12 Troubleshooting

Symptom	Explanation and action
SQL editor flags CASCADE or COMMENT	Check that the extension parses PostgreSQL, not SQL Server. Do not rewrite valid PostgreSQL merely to satisfy another dialect.
Loader rejects backslash commands	Use psql from the repository root; \copy and \set are client commands.
Validation reports a missing view	Create views after loading and before validation.
Power BI cannot authenticate	Use the local PostgreSQL database credentials and check the server and database names.
Reserved table name Measures	The final model uses Report Measures; do not rename it to the reserved word Measures.
Earlier date selection shows no operations	Check coverage; the contextual calendar starts earlier than the operational facts.
Financial value does not change with date	The financial sources are undated and intentionally have no calendar relationship.
Estimated ratios ignore region or airline type	These attributes are absent from the estimate source; use applicable airline or country selections.

13 Security and release rules

Publish the four business and technical PDF documents, source code, permitted data with attribution, PBIP source, diagrams, and final report assets. Keep editable Word sources in the ignored archive. Keep private guides, historical documents, audit logs, .venv, __pycache__, .pbi state, Office lock files, database backups, and credentials excluded. An ignore rule does not remove a file already tracked by Git or erase it from earlier commits.

Private records that were published in an earlier commit remain in that history unless a separate history-removal process is authorized. Do not rewrite history as part of documentation finalization. Enter passwords only into local credential prompts and replace any password previously shared in chat.

14 Technical boundaries

The supplied snapshot is static. Summary and estimate scope and methodology are not fully specified. No common flight identifier supports cross-source deduplication. Timestamp timezone is not supplied. Source date bounds do not prove complete observation. The report supports descriptive comparisons and planning discussion, not forecasting, causal attribution, or an operational risk probability.

Appendix A DAX measure catalog

Detailed Cancellations

COUNTROWS('aviation flight_cancellations')

Detailed Passengers Affected

SUM('aviation flight_cancellations'[passengers_affected])

Detailed Reroutes

COUNTROWS('aviation flight_reroutes')

Detailed Additional Distance Km

SUM('aviation flight_reroutes'[additional_distance_km])

Detailed Extra Fuel Cost USD

SUM('aviation flight_reroutes'[extra_fuel_cost_usd])

Detailed Reroute Delay Hours

SUM('aviation flight_reroutes'[delay_hours])

Modeled Summary Airline Loss USD

SUM('aviation vw_airline_loss_summary'[estimated_loss_usd])

Modeled Summary Cancellations

SUM('aviation vw_airline_loss_summary'[cancellations_count])

Modeled Summary Reroutes

SUM('aviation vw_airline_loss_summary'[reroutes_count])

Modeled Average Revenue Loss Pct

AVERAGE('aviation vw_airline_loss_summary'[revenue_loss_pct]) / 100

Estimated Daily Loss USD

SUM('aviation vw_airline_loss_estimate'[estimated_daily_loss_usd])

Estimated Cancellations

SUM('aviation vw_airline_loss_estimate'[cancelled_flights])

Estimated Reroutes

SUM('aviation vw_airline_loss_estimate'[rerouted_flights])

Estimated Fuel Cost USD

SUM('aviation vw_airline_loss_estimate'[additional_fuel_cost_usd])

Estimated Passengers

SUM('aviation vw_airline_loss_estimate'[passengers_impacted])

Estimated Cancellations per Estimated Reroute

DIVIDE([Estimated Cancellations], [Estimated Reroutes])

Estimated Passengers per Estimated Reroute

DIVIDE([Estimated Passengers], [Estimated Reroutes])

Estimated Fuel Cost per Estimated Reroute

DIVIDE([Estimated Fuel Cost USD], [Estimated Reroutes])

Airport Flights Affected

SUM('aviation vw_airport_disruption'[flights_affected])

Airport Disruption Records

COUNTROWS('aviation vw_airport_disruption')

Airport Disruption Hours

SUM('aviation vw_airport_disruption'[duration_hours])

Airspace Closure Records

COUNTROWS('aviation vw_airspace_closure')

Airspace Closure Hours

SUM('aviation vw_airspace_closure'[duration_hours])

Airspace Flights Affected

SUM('aviation vw_airspace_closure'[flights_affected])

Conflict Events

COUNTROWS('aviation conflict_events')

Military Event Records

COALESCE(CALCULATE([Conflict Events], 'aviation conflict_events'[event_type] = "Military"), 0)

Affected Flights by Factor

SWITCH(SELECTEDVALUE('Operational Factor'[Factor]), "Airport exposure", [Airport Flights Affected], "Airspace exposure", [Airspace Flights Affected])

Corridor Detailed Reroutes

SUM('aviation vw_route_passenger_impact'[rerouted_flights])

Corridor Additional Distance Km

SUM('aviation vw_route_passenger_impact'[additional_distance_km])

Corridor Extra Fuel Cost USD

SUM('aviation vw_route_passenger_impact'[additional_fuel_cost_usd])

Corridor Reroute Delay Hours

SUM('aviation vw_route_passenger_impact'[delay_hours])

Corridor Detailed Cancellations

SUM('aviation vw_route_passenger_impact'[cancelled_flights])

Corridor Cancellation Passengers

SUM('aviation vw_route_passenger_impact'[passengers_affected])

Appendix B Semantic relationship catalog

Dimension and key	Imported table and key	State
DimDate.Date	'aviation flight_cancellations'.date	Active
DimDate.Date	'aviation flight_reroutes'.date	Active
DimDate.Date	'aviation vw_airport_disruption'.date	Active
DimDate.Date	'aviation conflict_events'.date	Active
DimDate.Date	'aviation vw_airspace_closure'. 'Closure Start Date'	Active
DimDate.Date	'aviation vw_geopolitical_daily_impact'.date	Active

Dimension and key	Imported table and key	State
DimDate.Date	'aviation vw_route_passenger_impact'.date	Active
DimDate.Date	'aviation vw_airspace_closure'. 'Closure End Date'	Inactive
DimAirline.airline	'aviation vw_airline_loss_summary'.airline	Active
DimAirline.airline	'aviation vw_airline_loss_estimate'.airline	Active
DimAirline.airline	'aviation flight_cancellations'.airline	Active
DimAirline.airline	'aviation flight_reroutes'.airline	Active
DimCountry.country	'aviation vw_airline_loss_summary'.country	Active
DimCountry.country	'aviation vw_airline_loss_estimate'.country	Active
DimCountry.country	'aviation vw_airport_disruption'.country	Active
DimCountry.country	'aviation vw_airspace_closure'.country	Active
DimCountry.country	'aviation conflict_events'.country	Active
DimRegion.region	'aviation vw_airline_loss_summary'.region	Active
DimRegion.region	'aviation vw_airport_disruption'.region	Active
DimRegion.region	'aviation vw_airspace_closure'.region	Active
DimRegion.region	'aviation vw_regional_vulnerability'.region	Active
DimOriginCountry.origin_country	'aviation flight_cancellations'.origin_country	Active
DimOriginCountry.origin_country	'aviation flight_reroutes'.origin_country	Active
DimOriginCountry.origin_country	'aviation vw_route_passenger_impact'.origin_country	Active
DimDestinationCountry.destination_country	'aviation flight_cancellations'.destination_country	Active
DimDestinationCountry.destination_country	'aviation flight_reroutes'.destination_country	Active
DimDestinationCountry.destination_country	'aviation vw_route_passenger_impact'.destination_country	Active
DimOriginRegion.origin_region	'aviation flight_cancellations'.origin_region	Active

Dimension and key	Imported table and key	State
DimOriginRegion.origin_region	'aviation flight_reroutes'.origin_region	Active
DimOriginRegion.origin_region	'aviation vw_route_passenger_impact'.origin_region	Active
DimDestinationRegion.destination_region	'aviation flight_cancellations'.destination_region	Active
DimDestinationRegion.destination_region	'aviation flight_reroutes'.destination_region	Active
DimDestinationRegion.destination_region	'aviation vw_route_passenger_impact'.destination_region	Active
DimAirportSeverity.severity_level	'aviation vw_airport_disruption'.severity_level	Active
DimConflictSeverity.severity	'aviation conflict_events'.severity	Active

Appendix C Imported Power Query definitions

aviation conflict_events

```
let
    // Read the cleaned database records without changing their meaning.
    Source = PostgreSQL.Database("localhost", "aviation_bi"),
    Records = Source{[Schema="aviation",Item="conflict_events"]}[Data]
in
    Records
```

aviation flight_cancellations

```
let
    // Read the cleaned database records without changing their meaning.
    Source = PostgreSQL.Database("localhost", "aviation_bi"),
    Records = Source{[Schema="aviation",Item="flight_cancellations"]}[Data]
in
    Records
```

aviation flight_reroutes

```
let
    Source = PostgreSQL.Database("localhost", "aviation_bi"),
    aviation_flight_reroutes =
Source{[Schema="aviation",Item="flight_reroutes"]}[Data]
in
    aviation_flight_reroutes
```

aviation vw_airline_loss_estimate

```
let
  Source = PostgreSQL.Database("localhost", "aviation_bi"),
  Records = Source{[Schema="aviation",Item="vw_airline_loss_estimate"]}[Data]
in
  Records
```

aviation vw_airline_loss_summary

```
let
  Source = PostgreSQL.Database("localhost", "aviation_bi"),
  Records = Source{[Schema="aviation",Item="vw_airline_loss_summary"]}[Data]
in
  Records
```

aviation vw_airport_disruption

```
let
  Source = PostgreSQL.Database("localhost", "aviation_bi"),
  aviation_vw_airport_disruption =
Source{[Schema="aviation",Item="vw_airport_disruption"]}[Data]
in
  aviation_vw_airport_disruption
```

aviation vw_airspace_closure

```
let
  Source = PostgreSQL.Database("localhost", "aviation_bi"),
  aviation_vw_airspace_closure =
Source{[Schema="aviation",Item="vw_airspace_closure"]}[Data]
in
  aviation_vw_airspace_closure
```

aviation vw_geopolitical_daily_impact

```
let
  Source = PostgreSQL.Database("localhost", "aviation_bi"),
  aviation_vw_geopolitical_daily_impact =
Source{[Schema="aviation",Item="vw_geopolitical_daily_impact"]}[Data]
in
  aviation_vw_geopolitical_daily_impact
```

aviation vw_regional_vulnerability

```
let
  Source = PostgreSQL.Database("localhost", "aviation_bi"),
  aviation_vw_regional_vulnerability =
Source{[Schema="aviation",Item="vw_regional_vulnerability"]}[Data]
in
  aviation_vw_regional_vulnerability
```

aviation vw_route_passenger_impact

```
let
  Source = PostgreSQL.Database("localhost", "aviation_bi"),
  aviation_vw_route_passenger_impact =
Source{[Schema="aviation",Item="vw_route_passenger_impact"]}[Data]
in
  aviation_vw_route_passenger_impact
```

Appendix D PostgreSQL data dictionary

aviation airline_losses

Grain: One row per airline. Source: data/clean/airline_losses_clean.csv. Population: Undated modeled summary. Date scope: Undated.

Column	SQL type	Nullable	Unit or meaning
airline	TEXT	No	Source airline name
country	TEXT	No	Source or derived country label
airline_type	TEXT	No	Cargo, Flag Carrier, Low Cost, or Private
estimated_loss_usd	NUMERIC(15,2)	No	USD; supplied modeled total
cancellations_count	INTEGER	No	Modeled summary cancellation count
reroutes_count	INTEGER	No	Modeled summary reroute count
revenue_loss_pct	NUMERIC(5,2)	No	Percentage points 0 to 100
region	TEXT	No	Source region label

Checks: airline_type IN ('Cargo', 'Flag Carrier', 'Low Cost', 'Private'); estimated_loss_usd >= 0; cancellations_count >= 0; reroutes_count >= 0; revenue_loss_pct BETWEEN 0 AND 100.

aviation airline_losses_estimate

Grain: One row per airline. Source: data/clean/airline_losses_estimate_clean.csv. Population: Undated modeled estimate. Date scope: Undated.

Column	SQL type	Nullable	Unit or meaning
airline	TEXT	No	Source airline name
country	TEXT	No	Source or derived country label
estimated_daily_loss_usd	BIGINT	No	USD per estimated day; no reporting date
cancelled_flights	INTEGER	No	Estimated cancellation count
rerouted_flights	INTEGER	No	Estimated reroute count
additional_fuel_cost_usd	BIGINT	No	USD

Column	SQL type	Nullable	Unit or meaning
passengers_impacted	INTEGER	No	Estimated passenger exposure

Checks: estimated_daily_loss_usd >= 0; cancelled_flights >= 0; rerouted_flights >= 0; additional_fuel_cost_usd >= 0; passengers_impacted >= 0.

aviation airport_disruptions

Grain: One disruption record. Source: data/clean/airport_disruptions_clean.csv. Population: Simulated airport detail. Date scope: 2026-02-27 to 2026-06-07.

Column	SQL type	Nullable	Unit or meaning
airport_name	TEXT	No	Source airport name
iata_code	VARCHAR(3)	No	Three-character airport code
country	TEXT	No	Source or derived country label
region	TEXT	No	Source region label
disruption_type	TEXT	No	Source disruption category
severity_level	TEXT	No	Airport severity category
flights_affected	INTEGER	No	Reported affected-flight count; overlap unknown
duration_hours	NUMERIC(10,2)	No	Hours
date	DATE	No	Record date

Checks: char_length(iata_code) = 3; severity_level IN ('Low', 'Moderate', 'High', 'Severe', 'Critical'); flights_affected >= 0; duration_hours >= 0.

aviation airspace_closures

Grain: One closure record. Source: data/clean/airspace_closures_clean.csv. Population: Simulated closure detail. Date scope: 2026-02-28 to 2026-06-07.

Column	SQL type	Nullable	Unit or meaning
country	TEXT	No	Source or derived country label
region	TEXT	No	Source region label
closure_start_date	TIMESTAMP	No	Closure start timestamp; timezone not supplied

Column	SQL type	Nullable	Unit or meaning
closure_end_date	TIMESTAMP	No	Closure end timestamp; timezone not supplied
duration_hours	NUMERIC(10,2)	No	Hours
airspace_zone	TEXT	No	Source closure area label
reason	TEXT	No	Source reason label
flights_affected	INTEGER	No	Reported affected-flight count; overlap unknown

Checks: duration_hours >= 0; flights_affected >= 0; closure_end_date >= closure_start_date.

aviation conflict_events

Grain: One contextual event. Source: data/clean/conflict_events_clean.csv. Population: Simulated event context. Date scope: 2025-12-15 to 2026-06-08.

Column	SQL type	Nullable	Unit or meaning
date	DATE	No	Record date
event_type	TEXT	No	Source event category
event_description	TEXT	No	Source event description
source_location	TEXT	No	Source location after text trimming
location_name	TEXT	No	Text before first location comma
country	TEXT	Yes	Source or derived country label
severity	TEXT	No	Conflict severity category
aviation_impact	TEXT	No	Source impact description

Checks: severity IN ('Low', 'Medium', 'High', 'Very High', 'Critical').

aviation flight_cancellations

Grain: One cancellation record. Source: data/clean/flight_cancellations_clean.csv. Population: Simulated operational detail. Date scope: 2026-02-27 to 2026-06-07.

Column	SQL type	Nullable	Unit or meaning
date	DATE	No	Record date

Column	SQL type	Nullable	Unit or meaning
airline	TEXT	No	Source airline name
flight_number	TEXT	No	Source flight number; not a declared key
origin	TEXT	No	Origin airport label
destination	TEXT	No	Destination airport label
origin_country	TEXT	No	Source origin country
destination_country	TEXT	No	Source destination country
origin_region	TEXT	No	Source origin region
destination_region	TEXT	No	Source destination region
aircraft_type	TEXT	No	Source aircraft label
passengers_affected	INTEGER	No	Passengers in cancellation record
reason	TEXT	No	Source reason label

Checks: passengers_affected >= 0.

aviation flight_reroutes

Grain: One reroute record. Source: data/clean/flight_reroutes_clean.csv. Population: Simulated operational detail. Date scope: 2026-02-28 to 2026-06-07.

Column	SQL type	Nullable	Unit or meaning
date	DATE	No	Record date
airline	TEXT	No	Source airline name
flight_number	TEXT	No	Source flight number; not a declared key
origin	TEXT	No	Origin airport label
destination	TEXT	No	Destination airport label
origin_country	TEXT	No	Source origin country
destination_country	TEXT	No	Source destination country
origin_region	TEXT	No	Source origin region
destination_region	TEXT	No	Source destination region

Column	SQL type	Nullable	Unit or meaning
original_route	TEXT	No	Source original route text
new_route	TEXT	No	Source rerouted route text
original_distance_km	INTEGER	No	Kilometres
new_distance_km	INTEGER	No	Kilometres
additional_distance_km	INTEGER	No	Kilometres
extra_fuel_cost_usd	NUMERIC(12,2)	No	USD
delay_hours	NUMERIC(8,2)	No	Hours

Checks: original_distance_km >= 0; new_distance_km >= 0; additional_distance_km >= 0; extra_fuel_cost_usd >= 0; delay_hours >= 0; new_distance_km - original_distance_km = additional_distance_km.

Appendix E View lineage and output columns

vw_airline_loss_summary

Sources: airline_losses. Grain: One summary airline. Role: Authoritative summary; imported.

Semantic columns: airline, country, airline_type, region, estimated_loss_usd, cancellations_count, reroutes_count, revenue_loss_pct.

vw_airline_loss_estimate

Sources: airline_losses_estimate. Grain: One estimate airline. Role: Authoritative estimate; imported.

Semantic columns: airline, country, estimated_daily_loss_usd, cancelled_flights, rerouted_flights, additional_fuel_cost_usd, passengers_impacted.

vw_airport_disruption

Sources: airport_disruptions. Grain: One airport disruption record. Role: Supporting detail; imported.

Semantic columns: airport_name, iata_code, country, region, disruption_type, severity_level, flights_affected, duration_hours, date.

vw_airspace_closure

Sources: airspace_closures. Grain: One closure record. Role: Supporting detail; imported.

Semantic columns: country, region, closure_start_date, closure_end_date, duration_hours, airspace_zone, reason, flights_affected, Closure Start Date, Closure End Date.

vw_geopolitical_daily_impact

Sources: Conflict, airport, closure, cancellation, and reroute tables. Grain: One calendar date. Role: Supporting contextual baseline; imported.

Semantic columns: date, military_events, conflict_severity, airport_disruptions, airspace_closures, cancellations, reroutes, is_military_event_date.

Calendar rows: date, military_events, conflict_severity, airport_disruptions, airspace_closures, cancellations, reroutes, is_military_event_date. Daily operational counts represent supplied records; closure counts represent active calendar-date spans.

vw_regional_vulnerability

Sources: Airport, airspace, and modeled summary tables. Grain: One region. Role: Separate source aggregates; imported.

Semantic columns: region, airport_disruption_events, airports_affected, airport_flights_affected, airport_disruption_hours, estimated_loss_usd, airspace_closures, airspace_closure_hours, airspace_flights_affected.

vw_route_passenger_impact

Sources: Cancellation and reroute tables. Grain: Date plus origin/destination region and country. Role: Supporting route aggregates; imported.

Semantic columns: origin_region, destination_region, origin_country, destination_country, cancelled_flights, passengers_affected, rerouted_flights, additional_distance_km, additional_fuel_cost_usd, delay_hours, Corridor, date.

vw_airline_operational_financial

Sources: Summary and estimate tables. Grain: One union airline. Role: Deprecated compatibility; not imported; used by SQL validation.

Output combines summary fields, estimate fields, three estimate ratios, has_summary_data, and has_estimate_data. Missing source fields remain NULL. Source coverage is 73 union airlines.

vw_executive_kpi

Sources: Seven base tables. Grain: One summary row. Role: Deprecated compatibility; not imported.

Output is one row containing separate cancellation, passenger, reroute, distance, fuel, airport, airspace, financial, delay, conflict, and military controls. It is not a fact table for filtering.